

(2-3 Marks) LINEAR ALGEBRA

classmate

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★ Question from the Topics —

- i) Types of Matrix
 - ii) Determinants
 - iii) Rank of the Matrix
 - iv) Solution of linear Equations
 - v) Eigen Values and Eigen Vectors.
- } Most Important Topics.

Matrix —

Technique of storing the data.

$$M = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$\vec{a} = \hat{i} + 2\hat{j} + 4\hat{k} \quad \vec{b} = 2\hat{i} + 9\hat{j} + 5\hat{k} \quad \vec{c} = 4\hat{i} + 2\hat{j} + 8\hat{k}$$
$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 9 & 5 \\ 4 & 2 & 8 \end{bmatrix}$$

Row
Column
 $m \times n$
no. of Rows
no. of columns.
Order of the matrix.
(Size / dimension).

$M = (a_{ij})_{m \times n}$ → Position of the element
(having at the i th Row and j th column).

Type of Matrix — (Based on Size / dimension / order).

Rectangular Matrix
No. of Rows $\neq$ No. of Column
 $m \neq n$

Square Matrix.
No. of Rows = No. of Column
 $m = n$

Rectangular Matrices — ($m \neq n$).

i) Row Matrix — no. of Rows = 1 &
no. of Columns > 1
 $R = [1 \ 2 \ 3 \ 4]_{1 \times 4}$

Ex: $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$
 $R = [a_1 \ a_2 \ a_3]$

ii) Column Matrix — no. of Rows > 1 and
 no. of Column $= 1$

Ex: $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ 3×1 Ex: $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

iii) Zero Matrix or Null Matrix —

$a_{ij} = 0 \quad \forall \ i, j$

$O = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ 3×4

iv) Horizontal Matrix —
 no. of Rows (m) $<$ no. of Columns (n)

Ex: $H = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \end{bmatrix}$ 2×4

v) Vertical Matrix —

(m) no. of Rows $>$ no. of Columns (n)

Ex: $V = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$

Square Matrix — (no. of Rows = no. of Columns) ($m=n$)
 A Square Matrix is in form of —
 $M = (a_{ij})_{n \times n}$.

$$M = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}_{n \times n}$$

Main Diagonal, or Principle diagonal or leading diagonal.
 $i = j$

$$\sum_{i=1}^n a_{ii} = \text{Trace}$$

Trace = Sum of main diagonal elements.

$$\text{Trace} = a_{11} + a_{22} + a_{33} + \dots + a_{nn}$$

Basic Types of Square Matrix —

1. Diagonal Matrix — $A = (a_{ij})_{n \times n}$ in which $A = 0 \ \forall \ i \neq j$.

Lower Triangle $\rightarrow$ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$ $\rightarrow$ Upper Triangle $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ $\rightarrow$ Also a diagonal Matrix.

$\rightarrow$ Diagonal Matrix

2. Upper Triangular Matrix — $[A = 0, \ i > j]$

$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}_{n \times n}$ $\rightarrow$ Upper Triangle ($i < j$)

$\rightarrow$ Main diagonal ($i = j$)

Lower Triangle ($i > j$)

$\begin{bmatrix} 8 & 2 & 9 \\ 0 & 4 & 6 \\ 0 & 0 & 5 \end{bmatrix}$

iii) Lower Triangular Matrix —

$$A = (a_{ij})_{n \times n} \text{ where } a_{ij} = 0 \text{ for } i < j$$

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 6 & 4 & 0 \\ 8 & 2 & 5 \end{bmatrix} \rightarrow i < j$$

iv) Unit or Identity Matrix —

$$A = (a_{ij})_{n \times n} \text{ where}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$AI = A$$

v) Scalar Matrix —

$$A = [a_{ij}]_{n \times n} \text{ where}$$

$$A = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}$$

Algebra of Matrices —

1. Matrix Multiply by a Scalar — multiply each element of the matrix by scalar.

Ex:

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 1 \end{bmatrix}$$

$$4A = \begin{bmatrix} 4 & 8 & 16 \\ 8 & 12 & 4 \end{bmatrix}$$

$$V_1 = \hat{i} + 2\hat{j} + 4\hat{k}$$

$$V_2 = 2\hat{i} + 3\hat{j} + \hat{k}$$

$$4V_1 = 4\hat{i} + 8\hat{j} + 16\hat{k}$$

$$4V_2 = 8\hat{i} + 12\hat{j} + 4\hat{k}$$

Valid for both Rectangular and Square Matrix.

2. Addition of Matrices — Addition of corresponding elements of two Matrix.

$$\vec{U}_1 = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\vec{U}_2 = 3\hat{i} + 2\hat{j} + 5\hat{k}$$

$$R_1 = [1 \ 2 \ 3]_{1 \times 3}$$

$$R_2 = [3 \ 2 \ 5]_{1 \times 3}$$

$$\vec{U}_1 + \vec{U}_2 = (1+3)\hat{i} + (2+2)\hat{j} + (3+5)\hat{k}$$

$$\vec{U}_1 + \vec{U}_2 = 4\hat{i} + 4\hat{j} + 8\hat{k}$$

$$R_1 + R_2 = [4 \ 4 \ 8]_{1 \times 3}$$

Valid for both : Rectangular Matrix and Square Matrix
($m \times n$) ($n \times n$).

$$C_{m \times n} = [A]_{m \times n} + [B]_{m \times n} \quad C_{n \times n} = [A]_{n \times n} + [B]_{n \times n}$$

→ Add corresponding extremes.

Important Results —

i) Matrix Addition is commutative.

$$(A+B = B+A) = \text{Commutative.}$$

$$C_{m \times n} = [A]_{m \times n} + [B]_{m \times n} \text{ or}$$

$$C_{m \times n} = [B]_{m \times n} + [A]_{m \times n}.$$

ii) Matrix Addition is Associative.

$$A + (B+C) = A + (B+C) = (A+B) + C = \text{Associative.}$$

$$D_{m \times n} = [A]_{m \times n} + ([B]_{m \times n} + [C]_{m \times n}) \text{ or}$$

$$D_{m \times n} = ([A]_{m \times n} + [B]_{m \times n}) + [C]_{m \times n}.$$

iii) Cancellation Law for Matrix Addition.

$$A_{m \times n}, B_{m \times n}, C_{m \times n}$$

$$A_{m \times n} + B_{m \times n} = A_{m \times n} + C_{m \times n}$$

only when

$$B_{m \times n} = C_{m \times n}$$

i) Same Order

ii) All corresponding element of the matrix B and C are equal.

3. Matrix Multiplication —

$$A_{m \times n} \times B_{n \times m} = C_{m \times m} \quad (\text{Row Column Multiplication})$$

Condition: First Matrix = Second matrix
(no. of column) (no. of row).

Ex: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$ $B = \begin{bmatrix} 1 & 2 \end{bmatrix}_{1 \times 2}$

$C = AB = \text{not defined}$
no. of column $\neq$ no. of row.

Ex: $A = \begin{bmatrix} 1 & 2 \end{bmatrix}_{1 \times 2}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2}$

$C = AB = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ (No. of = No. of
 $AB = \begin{bmatrix} 7 & 10 \end{bmatrix}_{1 \times 2}$ Ans. Rows Columns).

$$A_{m \times n} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

$$B_{n \times m} = \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1m} \\ b_{21} & b_{22} & \dots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \dots & b_{nm} \end{bmatrix}$$

$$C_{m \times m} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1m} \\ c_{21} & c_{22} & \dots & c_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mm} \end{bmatrix}$$

$$C_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} + \dots + a_{1n}b_{n1}$$

no. of Column of 1st. no. of Row of 2nd.

$$C_{m \times n} = a_{m1}b_{1n} + a_{m2}b_{2n} + \dots + a_{mn}b_{nn}$$

Type 1: No. of multiplication required to findout $AB = C$.

No. of multiplication required to findout one element = n of matrix C .

Total No. of element in matrix = $m \times n$

Total no. of multiplication

required for $m \times n$ elements = $m \times n \times n$ elements.

Type 2: No. of Addition Required to findout $AB = C$.

2 elements added to form 1 element.

Similarly, 3 elements add to form 2 elements.

Hence, n elements added and no.

of addition required for n elements = $(n-1)$.

Total no. of elements in matrix $C = m \times n$.

Total no. of addition required to

findout C ($m \times n$) element = $m \times n \times (n-1)$.

Important Results —

i) Matrix Multiplication is not commutative.

$$A_{m \times n} B_{n \times m} = C_{m \times m}$$

$$B_{n \times m} A_{m \times n} = C \text{ (not defined)}$$

$$(AB \neq BA)$$

b. Matrix Multiplication is associative.

$$(AB)C = A(BC)$$

c. Matrix Multiplication follows distributive law.

$$A(B+C) = AB + AC$$

d. Matrix Multiple by Identity Matrix.

$$AI_n = I_n A = A$$

e. Positive Integer Power of Matrix.

$$A \cdot A = A^2$$

$$A^m \cdot A^n = A^{m+n}$$

A is a square Matrix.

Where m, n are Positive Integer.

Q. $A_{4 \times 5}$, $B_{5 \times 2}$, $C_{2 \times 1}$ Minimum no. of Multiplication required to find out ABC is 30.

Soln: Option 1: $A_{4 \times 5}$ $B_{5 \times 2}$
 $\swarrow \searrow$
 $D_{4 \times 2}$ No. of Multiplication required to find AB
 $= 4 \times 5 \times 2 = 40$

$$ABC = D_{4 \times 2} C_{2 \times 1} = \Delta$$

$$E_{4 \times 1} \rightarrow 4 \times 2 \times 1 = 8$$

$$48$$

Option 2: $B_{5 \times 2}$ $C_{2 \times 1}$

$$F_{5 \times 1} = 5 \times 2 \times 1 = 10$$

$$A_{4 \times 5} \quad F_{5 \times 1} \quad +$$

$$G_{4 \times 1} = 4 \times 5 \times 1 = 20$$

$$30$$

Hence, minimum no. of multiplication required is 30 - Ans

(Only for Square Matrix) Determinants (It can be positive, negative, fraction, zero, all variables).

Order 2×2 (Direct Method)
 Order 3×3 (General Method)
 Higher (order > 3) Order (General Method)

Determinants of 2×2 Matrix -

$$A_{2 \times 2} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$|A_{2 \times 2}| = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$|A_{2 \times 2}| = ad - bc$$

Product of Main diagonal

Product of Secondary diagonal.

Example: $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} = 1 \times 4 - 2 \times 3$
 $= 4 - 6 = -2$ Ans.

Determinants of 3×3 Matrix -

Ex: $A_{3 \times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 6 \\ 2 & 1 & 3 \end{bmatrix}$

General

Method: $1(15-6) - 2(3-12) + 3(1-10)$

$$9 - 2 \times (-9) + 3(-9)$$

$$9 + 18 - 27$$

0 - Ans.

Short Trick:

 $A_{3 \times 3} =$

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 5 & 6 \\ 2 & 1 & 3 \end{vmatrix}$$

→ Secondary Diagonal.

Principal diagonal

Parallel to
main diagonal elementParallel to
secondary diagonal. $1 \times 5 \times 3 \rightarrow 2 \times 6 \times 2$

$$(15 + 24 + 3) - (30 + 6 + 6) = 42 - 42 = 0 \text{ Ans.}$$

 $\rightarrow 1 \times 1 \times 3$

Determinants of 4×4 matrix — (By general method)
can be calculated same as 3×3 matrix.

Minor —

The determinant of square sub matrix obtained by deleting i^{th} row and j^{th} column. (M_{ij})

$$A_{3 \times 3} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 6 \\ 2 & 1 & 3 \end{bmatrix}$$

$$M_{11} = \begin{bmatrix} 5 & 6 \\ 1 & 3 \end{bmatrix} = 5 \times 3 - 1 \times 6 = 9.$$

 $M_{3 \times 3} =$

$$\begin{bmatrix} 9 & -9 & -9 \\ 3 & -3 & -3 \\ -3 & 3 & 3 \end{bmatrix}$$

→ Become negative.

Cofactor —

$$\text{Signed Minor} = (-1)^{i+j} M_{ij}$$

So, $i+j = \text{even}$ (remain same) $i+j = \text{odd}$ (become negative).

$$\text{Cofactor } 3 \times 3 = \begin{bmatrix} 9 & -9 & -9 \\ -3 & -3 & 3 \\ -3 & -3 & 3 \end{bmatrix}$$

Adjoint —

Transpose of the Cofactor Matrix.

Interchange of Row and Column of the Matrix.

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$

$$A^T = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

$$\text{So, } A^T = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

for Adjoint $\rightarrow$ Minor $\rightarrow$ Cofactor $\rightarrow$ Cofactor Transpose.

Shortcut:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 6 \\ 2 & 1 & 3 \end{bmatrix}$$

$$\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 2 & 3 \\ 1 & 5 & 6 & \times & \times & \times \\ 2 & 1 & 3 & \times & \times & \times \\ 1 & 2 & 3 & 1 & 2 & 3 \\ 1 & 5 & 6 & \times & \times & \times \\ 2 & 1 & 3 & \times & \times & \times \end{array}$$

$$\text{Adjoint: } \begin{bmatrix} 9 & -3 & -3 \\ 9 & -3 & -3 \\ -9 & 3 & 3 \end{bmatrix}$$

GATE

Q.

$$A = \begin{bmatrix} 2 & 3 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 2 \end{bmatrix} \text{ find Adjoint.}$$

Soln:

$$\begin{array}{ccc|ccc} 2 & 3 & 1 & 2 & 3 & 1 \\ 1 & -1 & 2 & \times & \times & \times \\ 3 & 1 & 2 & \times & \times & \times \\ 2 & 3 & 1 & 2 & 3 & 1 \\ 1 & -1 & 2 & \times & \times & \times \\ 3 & 1 & 2 & \times & \times & \times \end{array}$$

$$\text{Adjoint} = \begin{bmatrix} -4 & -5 & 7 \\ 4 & 1 & -3 \\ 4 & 7 & -5 \end{bmatrix}$$

Adjoint $\rightarrow$ Cofactor^T
 Cofactor^T $\rightarrow$ Adjoint (By Transpose).

Adjoint of Higher Order Matrix —

Adjoint $\rightarrow$ Matrix $\rightarrow$ option $\rightarrow$ element diff in all the matrix (A_{ij})

Example:
$$\text{Adjoint} = \begin{bmatrix} -6 & -11 & 7 \\ 2 & 5 & -3 \\ 4 & -7 & -5 \end{bmatrix}$$
 & Matrix $= \begin{bmatrix} 2 & 3 & 1 \\ 1 & -1 & 2 \\ 3 & 1 & 4 \end{bmatrix}$ (Cofactor (A_{ji})).

Adjoint $= (A^T)_{3 \times 2} \rightarrow ((2 \times 3) = (-1)^{2+3} M_{23})$

$$= (-1) \begin{vmatrix} 2 & 3 \\ 3 & 1 \end{vmatrix}$$

$$= (-1)(-7) = 7 \text{ Ans.}$$

Satisfied.

Laplace Expansion —

1. Row Expansion $\rightarrow$

Eg:
$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 5 & 6 \\ 2 & 1 & 3 \end{vmatrix} = 1(15-6) - 2(3-12) + 3(1-10)$$

$$= 9 + 18 - 27$$

$$= 27 - 27 = 0 \text{ Ans.}$$

2. Column Expansion —

Eg:
$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 5 & 6 \\ 2 & 1 & 3 \end{vmatrix} = 1(15-6) - 1(6-3) + 2(12-15)$$

$$= 9 - 3 - 6$$

$$= 0 \text{ Ans.}$$

Important Results of Determinants —

1. Eg: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$

$|A| = -2$ and also, $|A^T| = -2$
 So, $|A| = |A^T|$ ***

2. No. of Terms in the expansion of the determinant $(A_{n \times n}) = n!$ ***
 Proof: for 2×2 matrix -

$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \therefore$ No. of terms = 2 or $2!$.

for 3×3 matrix -

$A = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - gf) + c(dh - eg)$
 $= aei - afh - bdi + bgf + cdh - ceg$
 $\therefore$ No. of terms = 6 or $3!$.

3. $\begin{vmatrix} 0 & 0 & 0 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} \rightarrow |A| = 0$, $\begin{vmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{vmatrix} \rightarrow |A| = 0$, $\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 9 & 13 \end{vmatrix} \rightarrow |A| = 0$.

because of Zero row/
Zero Column.

because of Identical row/
Identical Column.

because 1st and 2nd rows
Proportional. (Row/column).

Determinant of the matrix = Zero.

4. $A = \begin{vmatrix} 1 & 2 & -3 \\ -4 & 1 & 3 \\ 2 & 0 & -2 \end{vmatrix} \rightarrow$ Sum = 0
 $\rightarrow$ Sum = 0
 $\rightarrow$ Sum = 0

If the elements of each row/
each column add up to
zero, then the $|A| = 0$ ***

5. $\begin{vmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{vmatrix}$, $\begin{vmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 5 & 4 & 2 \end{vmatrix}$, $\begin{vmatrix} 1 & 2 & 4 \\ 0 & 6 & 2 \\ 0 & 0 & 3 \end{vmatrix}$
 Δ diagonal Matrix $1 \times 8 \times 4 = 32$ Δ lower triangular matrix $1 \times 3 \times 2 = 6$ Δ upper triangular matrix $1 \times 6 \times 3 = 18$

Product of main diagonal element.

6. $|I_n| = 1.$

7.

$$A \xrightarrow{R_i \leftrightarrow R_j} B$$

$$|A| = -|B|$$

eg:- $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \rightarrow B = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$

$$|A| = 4 - 6 = -2$$

$$|B| = 6 - 4 = 2$$

$$|B| = -|A|$$

8.

$$A \xrightarrow{R_i \rightarrow kR_i} B$$

then

$$|B| = k|A|$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \xrightarrow{R_1 \rightarrow 3R_1} B = \begin{bmatrix} 3 & 6 \\ 3 & 4 \end{bmatrix}$$

$$|A| = -2$$

$$|B| = 12 - 18 = -6$$

9.

$$A \xrightarrow{R_i \rightarrow R_i + kR_j} B$$

$$|A| = |B|$$

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$$|A| = -2$$

$$R_1 \rightarrow R_1 + 2R_2$$

$$B = \begin{bmatrix} 7 & 10 \\ 3 & 4 \end{bmatrix}$$

$$|B| = -2$$

10.

$A_{n \times n}$,

$$|KA| = K^n |A|$$

Example:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$|A| = -2$$

$$3A = \begin{bmatrix} 3 & 6 \\ 9 & 12 \end{bmatrix}$$

$$|3A| = -18$$

$$= 9 \times (-2)$$

$$= (3)^2 \times |A|$$

11. $|AB| = |A||B|$
 12. $|A+B| \neq |A|+|B|$

Q. If $A = \begin{vmatrix} 1 & -2 & -1 \\ 2 & 3 & 1 \\ 0 & 5 & -2 \end{vmatrix}$ and $\text{Adj}(A) = \begin{vmatrix} -11 & -9 & 1 \\ 4 & -2 & -3 \\ 10 & K & 7 \end{vmatrix}$ then $K = \underline{-5}$.

Soln:

$$\begin{vmatrix} 1 & -2 & -1 & 1 & -2 \\ 2 & 3 & 1 & 2 & 3 \\ 0 & 5 & -2 & 0 & 5 \\ 1 & -2 & -1 & 1 & -2 \\ 2 & 3 & 1 & 2 & 3 \end{vmatrix}$$

$$\therefore \text{Adj}(A) = \begin{bmatrix} -11 & -9 & 1 \\ 4 & -2 & -3 \\ 10 & -5 & 7 \end{bmatrix}$$

Hence $K = -5$ Ans.

Q. The value of following determinant $\begin{vmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{vmatrix}$ is $\underline{-8}$. a. 8 c. -12 b. 12 d. -8.

Soln: $\begin{vmatrix} 1 & 4 & 9 \\ 4 & 9 & 16 \\ 9 & 16 & 25 \end{vmatrix}$ $C_3 \rightarrow C_3 - C_2$ $\begin{vmatrix} 1 & 3 & 5 \\ 4 & 5 & 7 \\ 9 & 7 & 9 \end{vmatrix}$ $C_3 \rightarrow C_3 - C_2$

$\begin{vmatrix} 1 & 3 & 2 \\ 4 & 5 & 2 \\ 9 & 7 & 2 \end{vmatrix}$ $R_3 \rightarrow R_3 - R_2$ $\begin{vmatrix} 1 & 3 & 2 \\ 4 & 5 & 2 \\ 5 & 2 & 0 \end{vmatrix}$ $\therefore 2(6-10)$
 $R_2 \rightarrow R_2 - R_1$ $\begin{vmatrix} 1 & 3 & 2 \\ 3 & 2 & 0 \\ 5 & 2 & 0 \end{vmatrix}$ $2 \times (-4)$
 -8 Ans.

Q. If a matrix A is $m \times n$ and B is $n \times p$, the no. of multiplication operation and addition operations needed to calculate AB are -
 1. mpn^2 , mpn

Soln: no. of multiplication required = mpn no. of addition required = $mp(n-1)$
 2. mpn , $(n-1)$
 3. mpn , $mp(n-1)$
 4. mpn^2 , $(mp)n$.

Q. The matrices $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ & $\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$ commute under multiplication -

1. If $a = b$ or $\theta = n\pi$ 3. Never
 2. Always 4. $a \cos \theta \neq b \sin \theta$.

Soln:

$$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\begin{bmatrix} a \cos \theta & -b \sin \theta \\ a \sin \theta & b \cos \theta \end{bmatrix} = \begin{bmatrix} a \cos \theta & -a \sin \theta \\ b \sin \theta & b \cos \theta \end{bmatrix}$$

$$a \sin \theta = b \sin \theta$$

$$(a-b) \sin \theta = 0$$

$$a = b \quad \text{or} \quad \sin \theta = 0 \Rightarrow \theta = n\pi$$

Q. The determinant of the matrix

$$\begin{vmatrix} 6 & -8 & 1 & 1 \\ 0 & 2 & 4 & 6 \\ 0 & 0 & 4 & 8 \\ 0 & 0 & 0 & -1 \end{vmatrix} \quad \begin{matrix} 9.11 & 3.0 \\ 48 & 4.24 \end{matrix}$$

Soln:

$$\begin{vmatrix} 6 & 2 & 4 & 6 \\ 0 & 4 & 8 \\ 0 & 0 & -1 \end{vmatrix}$$

OR

It is upper triangular matrix so,
 $|A| = 6 \times 2 \times 4 \times -1 = -48$ Ans.

$$6 [2(-4)] = 6[-8] = -48 \text{ Ans.}$$

Q.

- If A and B are two matrices and if AB exists then BA exists.
- i) Only if A has as many row as B has columns.
 - ii) only if both A and B are square matrices.
 - iii) Only if A and B are skew symmetric matrices.
 - iv) Only if both A and B are symmetric.

$$A_{m \times n} \quad B_{n \times p}$$

no. of column of A = no. of row of B.

$$B_{n \times p} \quad A_{m \times n}$$

$$[m \geq p]$$

no. of row of A = no. of column of B.

Inverse of the Matrix -

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

but, $|A| \neq 0$.

If $|A| = 0$ called non invertible Matrix or (Singular Matrix) and $|A| \neq 0$ called Invertible Matrix or (non Singular Matrix).

Note:-

Find out the inverse of the matrix:

Cofactor $\rightarrow$ Cofactor^T = Adjoint

Adjoint = Inverse.
 $|A|$

Find out the value of unknown element

Singular
 $|A| = 0$

Non Singular
 $|A| \neq 0$

Important Results -

1. $A_{2 \times 2}$ Matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Adjoint
Determinant

2. 3×3 Matrix $[A]$:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{-3} \begin{bmatrix} 5 & -2 & -3 \\ -4 & 1 & 6 \\ 0 & 0 & -3 \end{bmatrix}$$

3.

$$\text{Adjoint } A = A^{-1} |A|$$

4.

If $A_{n \times n}$:

$$|A^{-1}| = \frac{1}{|A|}$$

$$|\text{adjoint } A| = |A^{-1} |A||$$

$$|A|^n |A^{-1}|$$

$$\frac{|A|^n}{|A|} = |A|^{n-1}$$

5.

$$|\text{adjoint } A| = |A|^{n-1}$$

When A is of order $n \times n$.

6. $\text{adj adj } A = |A|^{n-2} \cdot A$ $n = \text{order of matrix } A$

7. $|\text{adj adj } A| = |A|^{(n-1)^2}$ $n = \text{order of matrix } A.$

Special Types of the Matrix -

1. **Symmetric Matrix** — A square matrix is said to be symmetric matrix if $A = A^T$ ***

$[a_{ij}]_{n \times n} = [a_{ji}]_{n \times n}.$
Eg: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 0 \\ 3 & 0 & 5 \end{bmatrix}$ Symmetric matrix is symmetric about main diagonal.

2. **Skew Symmetric Matrix** — A square matrix is said to be skew symmetric matrix if $A = -A^T$ ***

$[a_{ij}]_{n \times n} = -[a_{ji}]_{n \times n}.$

→ Main diagonal element of skew symmetric matrix is zero (0).
 $[a_{ii}]_{n \times n} = 0.$

→ $[a_{ij}]_{n \times n} + [a_{ji}]_{n \times n} + [0_{ii}]_{n \times n} = 0.$
Sum of all element of skew symmetric matrix is zero (0).

Eg: $A = \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & -4 \\ -3 & 4 & 0 \end{bmatrix}$

→ The determinant of skew symmetric matrix of odd order is zero.

Note:- Every square matrix can be expressed as sum of symmetric and skew symmetric matrix.

$$A = \underbrace{\frac{1}{2} [A+A^T]}_{\text{Symmetric}} + \underbrace{\frac{1}{2} [A-A^T]}_{\text{Skew Symmetric}}$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

$$A+A^T = \begin{bmatrix} 2a & b+c \\ b+c & 2d \end{bmatrix}$$

$$\frac{A+A^T}{2} = \begin{bmatrix} a & b+c/2 \\ b+c/2 & d \end{bmatrix}$$

$R = R^T = \Delta$ $\therefore R$ is Symmetric matrix.

$$R \text{ and } R^T = \begin{bmatrix} a & b+c/2 \\ b+c/2 & d \end{bmatrix}$$

Similarly another one is Skew Symmetric Matrix.

3. Orthogonal Matrix — Whenever, $A A^T = I$ or $A^T = A^{-1}$ then Matrix A is said to be Orthogonal Matrix.

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}, \quad A^T = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$A A^T = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$A A^T = \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & -\sin \theta \cos \theta + \sin \theta \cos \theta \\ -\sin \theta \cos \theta + \sin \theta \cos \theta & \sin^2 \theta + \cos^2 \theta \end{bmatrix}$$

$$A A^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \text{Identity Matrix.}$$

$\therefore$ A is orthogonal matrix then their Row Vectors (Column Vectors) form orthonormal set.

$$A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

$$X_1 = \begin{bmatrix} \cos \theta \\ -\sin \theta \end{bmatrix}, \quad X_2 = \begin{bmatrix} \sin \theta \\ \cos \theta \end{bmatrix}$$

$$\begin{aligned} \vec{X}_1 &= \cos \theta \hat{i} - \sin \theta \hat{j} \\ \vec{X}_2 &= \sin \theta \hat{i} + \cos \theta \hat{j} \end{aligned}$$

$$\vec{X}_1 \cdot \vec{X}_2 = \cos \theta \sin \theta - \sin \theta \cos \theta$$

$$\boxed{\vec{X}_1 \cdot \vec{X}_2 = 0}$$

Orthogonal.

→ Norms:- $|\vec{X}_1| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$
 $|\vec{X}_2| = \sqrt{\cos^2 \theta + \sin^2 \theta} = 1$

4. Idempotent Matrix — A square Matrix is said to be idempotent matrix if $\boxed{A^2 = A}$ ***

Example:- $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$

$$\therefore \boxed{A \cdot A = A^2 = A}$$

5. Nilpotent Matrix —

K = Positive least Integer.

$$\boxed{A^K = 0}$$

A is said to be nilpotent of index K.

Example: $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$

$$A^2 = A \cdot A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix} \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$$

$$A^2 = \begin{bmatrix} a^2b^2 - a^2b^2 & ab^3 - ab^3 \\ -a^3b + a^3b & -a^2b^2 + a^2b^2 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \therefore \boxed{A^2 = 0}$$

Complex Matrix —

(At least one 'i' term)

$$A = \begin{bmatrix} 1+2i & 3 \\ 4 & 5-2i \end{bmatrix}$$

★ Conjugate ($\bar{A}$) —

$$\bar{A} = \begin{bmatrix} 1-2i & 3 \\ 4 & 5+2i \end{bmatrix}$$

★ Conjugate Transpose ($\bar{A})^T$ —

$$(\bar{A})^T = \begin{bmatrix} 1-2i & 4 \\ 3 & 5+2i \end{bmatrix}$$

$$\therefore (\bar{A})^T = (\bar{A}^T)$$

$$A^T = \begin{bmatrix} 1+2i & 4 \\ 3 & 5-2i \end{bmatrix} \text{ and } (\bar{A}^T) = \begin{bmatrix} 1-2i & 4 \\ 3 & 5+2i \end{bmatrix}$$

1. Hermitian Matrix —

A complex Matrix A is Hermitian Matrix if $(\bar{A})^T = A$.

$$A = [a_{ij}]$$

$$A^T = [a_{ji}]$$

$$\text{ii)} \quad (\bar{A}^T) = [\bar{a}_{ji}] = [a_{ij}]$$

$$\text{iii)} \quad A_{n \times n} = \begin{bmatrix} a+ib & c \\ d+ei & f-ig \end{bmatrix} \quad \left(\begin{array}{l} \text{If diagonal element of hermitian} \\ \text{matrix is complex.} \end{array} \right)$$

for hermitian matrix,

$$(\bar{A})^T = A$$

$$\begin{bmatrix} a-ib & d-ei \\ c & f+ig \end{bmatrix} = \begin{bmatrix} a+ib & c \\ d+ei & f-ig \end{bmatrix}$$

$$a-ib = a+ib$$

$$f+ig = f-ig$$

It is only possible when $b=0$ and $g=0$.

And, Hence, the "principal diagonal is Real in Hermitian".

2. Skew Hermitian Matrix —

A complex Matrix A is Skew Hermitian Matrix if $(\bar{A})^T = -A$.

$$\text{ii)} \quad a_{ij} = -[\bar{a}_{ji}]$$

$$\text{iii)} \quad A = \begin{bmatrix} a+ib & c \\ d+ei & f \end{bmatrix}$$

$$(\bar{A})^T = \begin{bmatrix} a-ib & d+ei \\ c & f \end{bmatrix}$$

$$(\bar{A})^T = -A$$

$$\begin{bmatrix} a-ib & d-ei \\ c & f \end{bmatrix} = \begin{bmatrix} -a-ib & -c \\ -d-ie & -f \end{bmatrix}$$

for Skew Hermitian, matrix —

$$a-ib = -a-ib$$

$$a = -a$$

$$f = -f$$

$$f = -f$$

only when a and f is zero.

∴ Here Principal diagonal element of Skew Hermitian Matrix is "Zero" or "Purely Imaginary".

Note:- Every Square Matrix is expressed as sum of Hermitian and Skew Hermitian Matrix.

$$A = \frac{A + (\bar{A})^T}{2} + i \frac{A - (\bar{A})^T}{2i}$$

$$A = P + Qi$$

Hermitian
Matrix

Skew
Hermitian

3. Unitary Matrix —

A complex Matrix A is said to be unitary if

$$(\bar{A})^T A = I$$

or

$$(\bar{A})^T = A^{-1}$$

GATE Q. The inverse of the Matrix $A = \begin{bmatrix} -3 & 5 \\ 2 & 1 \end{bmatrix}$.

Soln:

$$A^{-1} = \frac{-1}{13} \begin{bmatrix} 1 & -5 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} -1/13 & 5/13 \\ +2/13 & +3/13 \end{bmatrix} \text{ Ans.}$$

GATE Q. If A and B are real Symmetric matrices of order n then which of the following is true.

i) $AA^T = I$ ✗

ii) $A = A^{-1}$ ✗

iii) $AB = BA$ ✗

iv) $(AB)^T = B^T A^T$

(Reversal law of Transpose).

Gate Q. Which of the following statement is not true?

i) $\det(AB) = \det(A) \det(B)$ (True)

ii) $\det(KA) = K^n \det(A)$ (True)

iii) $\det(A+B) = \det(A) + \det(B)$ (False).

iv) $\det(A) = 1 / \det(A^{-1})$.

$\det(A^{-1}) = 1 / \det(A)$. (True).

Gate Q. Let A be an invertible matrix and suppose that inverse of A is $\begin{bmatrix} -1 & 2 \\ 4 & -7 \end{bmatrix}$ then A .

Soln: $A^{-1} = \begin{bmatrix} -1 & 2 \\ 4 & -7 \end{bmatrix}$

$(A^{-1})^{-1} = A = \frac{1}{-1} \begin{bmatrix} -7 & -2 \\ -4 & -1 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 4 & 1 \end{bmatrix}$ Ans.

Gate Q. The matrix $A = A^T$, where A is any matrix is -

i) Skew Symmetric

iii) Always Symmetric

ii) Symmetric about secondary diagonal

iv) Another general matrix.

Soln: If $A = A^T$ then it is always symmetric.

Note: i) If A and B are symmetric matrix then $AB+BA$ is symmetric and $AB-BA$ is skew symmetric also if A and B are skew symmetric matrix then $AB+BA$ is symmetric and $AB-BA$ is skew symmetric.

ii) If A is symmetric then A^n is symmetric and if A is skew symmetric then A^{2n} is symmetric and A^{2n+1} is skew symmetric matrix.

Gate Q. If A and B are non zero square matrices, then $AB = 0$ implies,

i) A and B are orthogonal

iii) B is singular $\times$

ii) A and B are singular

iv) A is singular $\times$

Soln:

$$|AB| = 0$$

$$|A||B| = 0 \quad \text{means} \quad |A| = 0 \text{ or } |B| = 0 \text{ or } |A| \neq 0 \text{ and } |B| = 0.$$

Case I: Let A is singular and B is non singular.

$$|B| \neq 0 \rightarrow \text{Inverse exists}$$

$$AB = 0$$

$$AB B^{-1} = 0$$

$$|A| = 0 \quad (\text{X})$$

Case II: Let B is singular and A is non singular.

$$|A| \neq 0 \rightarrow \text{Inverse exists}$$

$$AB = 0$$

$$A A^{-1} B = 0$$

$$|B| = 0 \quad (\text{X})$$

Q. Let $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ and $B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$ be two matrices such that $AB = I$. Let $C = A \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $CD = I$. Express the elements of D in terms of the elements of B.

Soln:

$$CD = I$$

$$D = C^{-1}$$

$$\text{Note! } (AB)^{-1} = B^{-1}A^{-1}$$

$$D = \left[A \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \right]^{-1}$$

$$D = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}^{-1} A^{-1} \rightarrow \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} A^{-1}$$

$$D = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \quad \begin{matrix} AB = I \\ A^{-1} = B \end{matrix}$$

$$D = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} - b_{11} & b_{22} - b_{12} \end{bmatrix}$$

Q.

Let $A_{m \times n}$ be matrix of order n and I_n be the matrix obtained by interchanging the first and second rows of I_n . Then AI_n is such that its first -

- i) Row is the same as its second row.
 ii) Row is the same as second row of A.
 iii) Column is the same as the second column of A.
 iv) Row is a ^{zero} row.

Soln: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $I_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $I_2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
 $- AI_2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$

Exeg. If A is a real square Matrix, the AA^T is

- i) Unsymmetric iii) Skew Symmetric
 ii) Always Symmetric iv) Sometimes Symmetric.

Soln: $A = \text{Real Square Matrix}$
 $B = AA^T$

$$B^T = (AA^T)^T = (A^T)^T(A)^T = AA^T$$

$\boxed{B^T = B}$ Hence, it is always Symmetric.

Gate Q. If matrix Algebra $AS = AT$ (A, S, T are of appropriate order) implies $S = T$ only if -

- i) A is Symmetric iii) A is non Singular
 ii) A is Singular iv) A is Skew Symmetric.

Soln: $AS = AT$ (Multiplying by A^{-1} both sides)
 $A^{-1}AS = A^{-1}AT$

$$S = T$$

and $A^{-1} = \frac{\text{Adj } A}{|A|}$ $|A| \neq 0 \rightarrow A$ is non Singular Matrix.

Gate

Q. If $\Delta = \begin{bmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{bmatrix}$ then which of the following is a factor of Δ .

Q. a. $a+b$ b. $a-b$ c. abc d. $a+b+c$

Soln:

$$\Delta = \begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix}$$

$$\begin{aligned} \Delta &= \begin{vmatrix} 1 & a & bc \\ 0 & b-a & ca-bc \\ 0 & c-a & ab-ca \end{vmatrix} = 1 \{ (b-a)(ab-ca) - (c-b)(ca-bc) \} \\ &= a(b-a)(b-c) - c(c-b)(a-b) \\ &= c(b-c)(a-b) - a(b-c)(a-b) \\ \Delta &= (a-b)(b-c)(c-a). \end{aligned}$$

$(a-b)$ is a factor of Δ .

Q. If $A = \begin{vmatrix} 1 & -2 & -1 \\ 2 & 3 & 1 \\ 0 & 5 & -2 \end{vmatrix}$ and $\text{Adj}(A) = \begin{vmatrix} -11 & -9 & 1 \\ 4 & -2 & -3 \\ 10 & K & 7 \end{vmatrix}$ then $K =$

Soln:

$$A = \begin{vmatrix} 1 & -2 & -1 \\ 2 & 3 & 1 \\ 0 & 5 & -2 \end{vmatrix}$$

$$\text{Adj}(A) = \begin{vmatrix} -11 & -9 & 1 \\ 4 & -2 & -3 \\ 10 & -5 & 7 \end{vmatrix}$$

$$\begin{vmatrix} 1 & -2 & -1 & 1 & -2 \\ 2 & 3 & 1 & 2 & 3 \\ 0 & 5 & -2 & 0 & 5 \\ 1 & -2 & -1 & 1 & -2 \\ 2 & 3 & 1 & 2 & 3 \end{vmatrix}$$

$$\therefore K = -5$$

Q. If A is any $n \times n$ matrix and K is a scalar, $|KA| = a|A|$, where a is K^n .
 $|KA| = K^n |A|$
 On comparing $a = K^n$.

Q. The eqn $\begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & -1 \\ y & y^2 & y \end{vmatrix} = 0$, represents a parabola passing through the points.

i) $(0,1), (0,2), (0,-1)$

ii) $(0,0), (-1,1), (1,2)$

iii) $(1,1), (0,0), (2,2)$

iv) $(1,2), (2,1), (0,0)$

Soln:

$$\begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & -1 \\ y & x^2 & x \end{vmatrix} = 0$$

$$2(x + x^2) - 1(x + y) + 1(x^2 - y) = 0$$

$$2x + 2x^2 - x - y + x^2 - y = 0$$

$$3x^2 + x - 2y = 0$$

$(0,0)$, $(-1,1)$ and $(1,2)$ satisfies the above equation.

Rank of the Matrices —

Minor → Determinant of square sub. Matrix obtained by deleting i th row and j th column.

★ Rank → "Highest order non zero minor".

Example: $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & 3 & 5 & 7 \\ 2 & 4 & 6 & 8 \end{bmatrix}_{3 \times 4}$

★ Case I: 1×1 —

$|1|, |2|, |3|, |4|, |5|, |7|, |6|, |8|,$

★ Case II: 2×2 —

$$\begin{vmatrix} 1 & 2 \\ 1 & 3 \end{vmatrix}, \begin{vmatrix} 1 & 3 \\ 1 & 5 \end{vmatrix}, \begin{vmatrix} 1 & 4 \\ 1 & 7 \end{vmatrix}, \dots$$

$1 \neq 0$

★ Case III: 3×3 —

$$\begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 2 & 4 & 6 \end{vmatrix} = 0, \quad \begin{vmatrix} 1 & 2 & 4 \\ 1 & 3 & 7 \\ 2 & 4 & 8 \end{vmatrix} = 0$$

Rows are Proportional.

Rank → Highest order non zero Minor:

$$\begin{array}{l} \boxed{3 \times 3} \rightarrow \text{all are zero. (X)} \\ \boxed{2 \times 2} \rightarrow \text{Rank} = 2 \\ \boxed{1 \times 1} \end{array}$$

★ Rank — "The number of non zero Row in the Echelon form of the matrix"

● Elementary Row operation —

1. Interchanging of Two Rows.

$$R_i \leftrightarrow R_j$$

Eg:
$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{bmatrix} 9 & 10 & 11 & 12 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \end{bmatrix}$$

2. Multiply a row with non zero constant (K) —

$$R_i \rightarrow KR_i$$

Eg:
$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_2 \rightarrow 2R_2} \begin{bmatrix} 1 & 2 & 3 \\ 8 & 10 & 12 \\ 7 & 8 & 9 \end{bmatrix}$$

3. Adding a row multiple of the constant.

$$R_i \rightarrow R_i + KR_j$$

Eg:
$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 + R_2} \begin{bmatrix} 5 & 7 & 9 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

★ Echelon form of the matrix —

- i) Zero row [if any] must be at the bottom of the matrix.
- ii) The leading non zero element of the row [if any] must be right hand side of the previous row leading element.

Eg:
$$\begin{bmatrix} 4 & 8 & 5 & 6 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Zero row at the bottom.

∴ Echelon form.

Eg:
$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

leading non zero element = first non zero element.

→ Echelon form.

3x3

Eg: $\begin{bmatrix} (1.E) & & \\ & (1.E) & \\ \rightarrow \text{Zero} & \text{Zero} & \text{Zero} \end{bmatrix}_{3 \times 3} \rightarrow \text{Yes, it is in Echelon form.}$

Zero row
at the bottom

Eg: $\begin{bmatrix} (1.E) & & \\ & (1.E) & \\ & & (1.E) \end{bmatrix}_{3 \times 3}$

not in Right hand Side. So not
Echelon form.

Eg: $\begin{bmatrix} (1.E) & & & \\ & (1.E) & & \\ & & (1.E) & \\ & & & (1.E) \end{bmatrix}$

not in Echelon form.

Eg: $\begin{bmatrix} & & (1.E) \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Echelon form of Matrix.}$

Eg: $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \text{Echelon form.}$

Q. Convert into Echelon form -

$A = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 8 \\ 2 & 3 & 5 & 16 \end{bmatrix}$

Problem

$R_2 \rightarrow R_2 - R_1$
 $R_3 \rightarrow R_3 - 2R_1$

Soln:

~~$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 8 \\ 2 & 3 & 5 & 16 \end{bmatrix}$~~

$A = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \end{bmatrix}$

Problem.

$R_3 \rightarrow R_3 - R_2$

$A = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix}$

$\rightarrow$ This the Echelon form of Matrix.

Hence, the Rank of the Matrix is 3.

- Rank $\rightarrow$ Highest order non zero minor of the Matrix.
- $\rightarrow$ No. of non zero Row in the Echelon form of the matrix.

Q. The rank of the Matrix $M = \begin{bmatrix} 5 & 10 & 10 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}$ is _____.
 a. 1 b. 3 c. 2 d. 0.

Soln:

$$R_1 \rightarrow R_1 - R_3$$

$$\begin{bmatrix} 2 & 4 & 4 \\ 1 & 0 & 2 \\ 3 & 6 & 6 \end{bmatrix}$$

$$\text{Again, } R_3 \rightarrow R_3 - R_1$$

$$\begin{bmatrix} 2 & 4 & 4 \\ 1 & 0 & 2 \\ 1 & 2 & 2 \end{bmatrix}$$

$$R_1 \rightarrow R_1 - 2R_3$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 2 \\ 1 & 2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & 2 & 2 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Hence, Rank is 2.

Q. The rank of the Matrix, $M = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ is _____.
 a. 0 c. 2 b. 3 d. -1.

Soln:

$$M = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_1$$

$$R_3 \rightarrow R_2 + R_3$$

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$\therefore \text{Rank} = 3.$

Q. The rank of the matrix $\begin{bmatrix} -4 & 1 & -1 \\ -1 & -1 & -1 \\ 7 & -3 & 1 \end{bmatrix}$ is _____.
 a. 1 b. 2 c. 3 d. 4.

Soln: Method I:

$$\begin{vmatrix} -4 & 1 & -1 \\ -1 & -1 & -1 \\ 7 & -3 & 1 \end{vmatrix}$$

$$-4(-1-3) + 1(1-3) + 7(-1-1)$$

$$-4(-4) - 2 - 14 = 16 - 16 = 0$$

Rank cannot be 3.

$$\begin{vmatrix} -4 & 1 \\ -1 & -1 \end{vmatrix}$$

$$= 4 + 1 = 5 \neq 0$$

Hence, Rank = 2 Ans.

Q. The Rank of the matrix $\begin{bmatrix} 1 & 1 & 0 & -2 \\ 2 & 0 & 2 & 2 \\ 4 & 1 & 3 & 1 \end{bmatrix}$ a. 1 b. 3 c. 2 d. 4

Sol: $\begin{vmatrix} 1 & 1 & 0 \\ 2 & 0 & 2 \\ 4 & 1 & 3 \end{vmatrix} \rightarrow 1(0-2) - 1(6-8) = 1(0-2) - 1(-2) = 0$

$\begin{vmatrix} 1 & 0 & -2 \\ 2 & 2 & 2 \\ 4 & 3 & 1 \end{vmatrix} \rightarrow 1(2-6) - 2(6-8) = 1(-4) - 2(-2) = -4 + 4 = 0$

Hence, Rank = 2

Hence, Rank cannot be 3.

Important Results on Rank —

1. $R(A) = n$ means —

i) There is at least one non zero Minor of order 'n'.

ii) Minor of order greater than 'n' must be equal to zero.

2. $R(A_{m \times n}) \leq \min(m, n)$

Eg: $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{bmatrix}_{3 \times 4}$

Maximum Possible Rank = 3

3. $\text{Rank}(A) = \text{Rank}(A^T)$

$\xrightarrow{\text{b/z}} |A| = |A^T|$

4. $\text{Rank}(A) = 0 \rightarrow A \rightarrow$ is only when Zero matrix or null matrix.

5. If $A_{m \times n}, B_{n \times 1}$

$\text{Rank}(B_{n \times 1} A_{m \times n}) = 1$

Example: $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}_{1 \times 3}$ and $B = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}_{3 \times 1}$

$BA = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}_{3 \times 1} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix}_{1 \times 3} = \begin{bmatrix} 4 \times 1 & 4 \times 2 & 4 \times 3 \\ 5 \times 1 & 5 \times 2 & 5 \times 3 \\ 6 \times 1 & 6 \times 2 & 6 \times 3 \end{bmatrix}_{3 \times 3}$

Proportional

$$R_3 \rightarrow R_3 - \frac{3}{2} R_1$$

$$R_2 \rightarrow R_2 - \frac{5}{4} R_1$$

$$\begin{bmatrix} 4 \times 1 & 4 \times 2 & 4 \times 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \therefore \text{Rank} = 1.$$

6. Rank $(AB) \leq \min \{ \text{Rank}(A), \text{Rank}(B) \}.$

7. Rank $(A^T) = \text{Rank}(A).$

Q. Given Rank $(A) = 2$ then find Rank (A^T) ?

Soln:

Rank $(A^T) = 2$ Am.

8. If A is a square Matrix —
 $A_{n \times n}$

↓
Singular Matrix

Non Invertible $\Rightarrow |A| = 0$

Rank $(A) < n$

↓
Non Singular Matrix.

Invertible Matrix $\Rightarrow |A| \neq 0$

Rank $(A) = n$

Reduced Row Echelon form —

↓ use

System of Linear Equation + Eigen Vectors.

① $\rightarrow$ Echelon form.

②

③ leading non zero element of the row (if any) must be 1. (if any).

④ The leading non zero element of the row is the only non zero element in its column.

Example:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

① and ② Satisfied

③ Satisfied

④ Satisfied.

$\therefore$ Reduced Row Echelon form.

Example :

$$\left[\begin{array}{ccc|cc} 1 & 0 & 1 & 0 & \\ 0 & 1 & 5 & 0 & \\ 0 & 0 & 0 & 1 & \end{array} \right]$$

(i) $\rightarrow$ No Zero Row $\rightarrow$ Satisfied

(ii) $\rightarrow$ Satisfied

(iii) $\rightarrow$ leading element 1 $\rightarrow$ Satisfied.

(iv) $\rightarrow$ Satisfied. (Bcoz third column doesn't have leading element).

$\therefore$ Reduced Row Echelon form.

Example :

$$\left[\begin{array}{ccc|cc} 1 & 1 & 1 & & \\ 0 & 0 & 0 & & \\ 0 & 0 & 0 & & \end{array} \right]$$

I $\rightarrow$ Satisfied

II $\rightarrow$ Satisfied

III $\rightarrow$ Satisfied

IV $\rightarrow$ Satisfied

$\therefore$ Reduced Row Echelon form.

Example :

$$\left[\begin{array}{ccc|cc} 0 & 0 & 0 & & \\ 0 & 0 & 0 & & \\ 0 & 0 & 0 & & \end{array} \right]$$

I $\rightarrow$ Satisfied

II $\rightarrow$ Satisfied

III $\rightarrow$ Satisfied

IV $\rightarrow$ Satisfied

$\therefore$ Reduced Row Echelon form.

★

Matrix

Elementary Row Transformation

Echelon form

Elementary Row Transformation

Reduced Row Echelon form

Q. Convert into reduced row Echelon form :

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 8 \\ 2 & 3 & 5 & 16 \end{array} \right]$$

Soln:

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 3 & 4 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$\therefore$ Echelon form.

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 4 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$R_1 \rightarrow R_1 + R_3$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

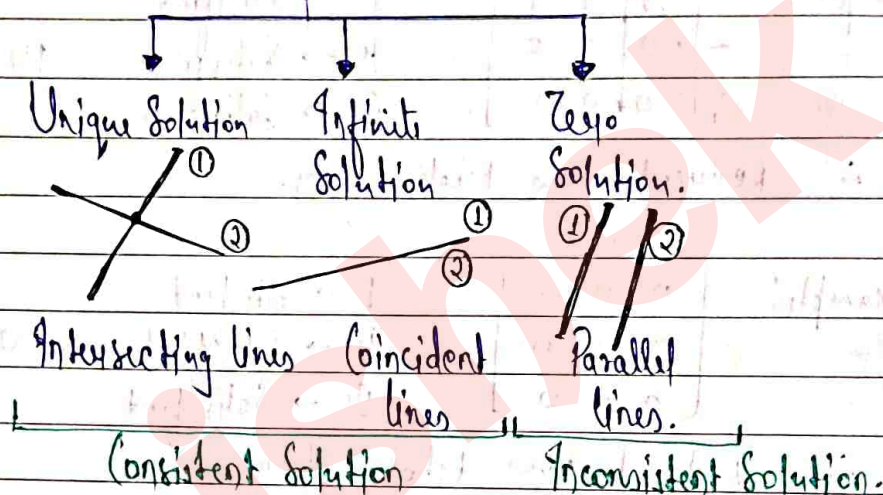
$\therefore$ Reduced Row Echelon form.

System of Linear Equations —

Types of Solution —

$$a_1x + b_1y = c_1$$

$$a_2x + b_2y = c_2$$



$$AX = B$$



Homogeneous linear system
(Same throughout) $B = 0$

Non Homogeneous linear system
 $B \neq 0$

• Augmented Matrix :- No Significance.

$$\begin{bmatrix} A & | & B \end{bmatrix}$$

★ Non Homogeneous linear system —

$n = \text{no. of unknown}$

Inconsistent Solution
(No Solution)

Consistent Solution.

$$R(A) = R(A|B)$$

$$R(A) \neq R(A|B)$$

Unique Solⁿ

$$R(A) = n$$

Infinite Solⁿ

$$R(A) < n$$

No. of linearly Independent Solution or
No. of Free Variable or
Dimension of Solution space = $n - m$

Q. The System $\begin{cases} x+y+z=6 \\ x+4y+6z=20 \\ x+4y+dz=14 \end{cases}$
Non Homogeneous linear System.

Find the value of d and 14 , so that the system has.

- a. Unique Soln
b. Infinite Soln and c. No. Solution.

Soln: $n = \text{no. of variables} = 3$
 $m = \text{no. of equation} = 3$ } Determined linear System.

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 4 & 6 \\ 1 & 4 & d \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - R_1]{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 5 \\ 0 & 3 & d-1 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 5 \\ 0 & 0 & d-6 \end{bmatrix}$$

$$[A:B] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 4 & 6 & 20 \\ 1 & 4 & d & 14 \end{array} \right] \xrightarrow[R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1]{AB} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 3 & 5 & 14 \\ 0 & 3 & d-1 & d-6 \end{array} \right]$$

$$R_3 \rightarrow R_3 - R_2 \rightarrow \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 3 & 5 & 14 \\ 0 & 0 & d-6 & d-20 \end{bmatrix} \xrightarrow{\text{E.f of } A} \text{E.f of } AB.$$

i) Zero Solution.

$$\text{Rank}(A) \neq \text{Rank}(A:B)$$

Ans = If $d=6$ then $\text{Rank}(A)=2$ and $14 \neq 20$ then $\text{Rank}(A:B)=3$.
and If $d=6$ then $\text{Rank}(A)=2$ and $14=20$ then $\text{Rank}(A:B)=2$,
which is not Possible.

ii) Infinite Soln -

$$\boxed{\text{Rank}(A) = \text{Rank}(A:B) = M} < n$$

If $M = 6$ and $n = 20$ then

$$\text{Rank}(A) = \text{Rank}(A:B) = 2$$

3

iii) Unique Soln -

$$\boxed{\text{Rank}(A) = \text{Rank}(A:B) = M} = n$$

If $M \neq 6$

$$\text{Rank}(A) = 3$$

Independent of M becoz 3

$$M = 20$$

$$M \neq 20$$

$$\text{Rank}(A:B) = 3$$

Type of Solution

$$\text{Rank}(A) = \text{Rank}(A:B) = n$$

Then → Unique Solution.

* Unique Solution

Find the value of constant x and y for unique soln.

$$\text{Echelon form} = \text{Rank}(A) = \text{Rank}(A:B) = n$$

Find out the value of variable for Unique Soln → Reduced Row Echelon.

Q.

$$x + y + z = 6$$

$$x + 2y + 3z = 8$$

$$2x + 3y + 6z = 18$$

Non homogeneous linear system.

Solution: Types of Solution -

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 8 \\ 2 & 3 & 6 & 18 \end{array} \right]$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 4 & 6 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 2 & 4 \end{array} \right] \rightarrow A:B$$

$$\therefore \text{Rank}(A) = \text{Rank}(A:B) = n = 3$$

Unique Soln.

Rank(A) = 3 and
Rank(A:B) = 3 and $n = 3$.
 $n = \text{no. of variables}$.

Solution $\rightarrow$ $x + y + z = 6$
(from Echelon form). $y + 2z = 2$
 $2z = 4$

$$[z = 2]$$

$$y + 2z = 2$$

$$y + 4 = 2$$

$$[y = -2]$$

$$x + y + z = 6$$

$$x - 2 + 2 = 6$$

$$[x = 6]$$

Extra

Solution $\rightarrow$
(from reduced row echelon form).

$$\left[\begin{array}{ccc|c} \textcircled{1} & 1 & 1 & 6 \\ 0 & \textcircled{1} & 2 & 2 \\ 0 & 0 & \textcircled{2} & 4 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3/2} \left[\begin{array}{ccc|c} \textcircled{1} & 1 & 1 & 6 \\ 0 & \textcircled{1} & 2 & 2 \\ 0 & 0 & \textcircled{1} & 2 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 4 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 2 \end{array} \right] \xrightarrow{\begin{array}{l} R_1 \rightarrow R_1 + R_3 \\ R_2 \rightarrow R_2 - 2R_3 \end{array}} \left[\begin{array}{ccc|c} \textcircled{1} & 0 & 0 & 6 \\ 0 & \textcircled{1} & 0 & -2 \\ 0 & 0 & \textcircled{1} & 2 \end{array} \right] \text{ Reduced$$

$$[x = 6] \text{ and } [y = -2] \text{ and } [z = 2] \rightarrow \text{Row Echelon form.}$$

Types of Solution: $\text{Rank}(A) \neq \text{Rank}(A:B)$

★ Zero Solution

$\rightarrow$ Find out the value for variable for Zero Solution. (Find Echelon form then find the value of x, y and z).

$\rightarrow$ Find the value of 1 and 4 for Zero Solution.

$$\rightarrow \text{E. form} = \text{Rank}(A) \neq \text{Rank}(A:B)$$

- ★ Infinite Solution —
- Types of Solution.
 - $\text{Rank}(A) = \text{Rank}(A:B) < n$.
 - Find the values of r and n for Infinite Solution.
 - E.g. $\rightarrow \text{Rank}(A) = \text{Rank}(A:B) < n$.
 - Find the no. of free variables / basic variables.
 - Linearly independent Solution or Dimension of Solⁿ space $= n - r$.
 - Find out linearly Independent Vectors.

Q. $x + y + z = 6$
 $x + 2y + 3z = 8$
 $2x + 3y + 4z = 14$

Soln: • Type 1: Types of Solution.

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 8 \\ 2 & 3 & 4 & 14 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - R_1$$

$$R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 1 & 2 & 2 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank $[A:B] = 2$

$$\text{Rank}(A) = 2$$

$$\therefore 2 = \text{Rank}(A) = \text{Rank}[A:B] < 3.$$

Hence, Infinite Solution.

- Type 2: no. of L.I vectors or no. of free variables or Dimension of Solⁿ space $= n - r = 3 - 2 = 1$.

Reduced Row Echelon form:

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 4 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Reduced Row
Echelon form.

★ Basic Variables —

Variable corresponding to leading non zero element in Reduced Row Echelon form.

$$x, y. (2) = \text{rank.}$$

★ Free Variables —

Total no. of variables — Basic Variables

$$3 - 2 = 1.$$

$$x - z = 4$$

then

$$x = 4 + z$$

$$y + 2z = 2$$

$$y = 2 - 2z$$

$$\text{let, } z = t$$

$$x = 4 + t \text{ and } y = 2 - 2t$$

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4+t \\ 2-2t \\ t \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix} + t \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

Linearly
Independent
Vectors.

Homogeneous Linear System — (Consistent Solution).

$$AX = B \quad (B \neq 0)$$

$$AX = 0$$

Unique Solution

$$[M = n]$$

(Trivial Solution)

Infinite Solution

$$[M < n]$$

(Non Trivial Solution).

$$AX = 0$$



Square Matrix

Unique Soln (Trivial Solution)

Infinite Solution (Non Trivial Solution)

$$|A| \neq 0$$

$$|A| = 0$$

→ Highest order non zero Minor (Rank = n).

2. If $AX = 0$ has infinite solution.

No. of L.I vector or

No. of free Variable or $= n - m$

Dimension of free variable or Nullity

Q. The number of linear independent solution of the system of equations.

$$\begin{bmatrix} 1 & 0 & 2 \\ 1 & -1 & 0 \\ 2 & -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0 \text{ is equal to } \underline{\quad} \text{ a. 1 c. 3} \\ \text{b. 2 d. 0.}$$

Soln:

Homogeneous linear system

$$AX = 0$$

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & -1 & 0 \\ 2 & -2 & 0 \end{bmatrix}$$

$$|A| = 1(0-0) + 2(-2+2)$$

$$|A| = 0$$

∴ Infinite solution.

Rank $\neq 3$ becoz $|A| = 0$ and hence Rank = 2.

∴ No. of linearly independent solution = $n - m$
 $= 3 - 2 = 1$ Ans.

Q. Solve the following system of equations.

$$x_1 + x_2 + x_3 = 3$$

a. Unique Solution

$$x_1 - x_2 = 0$$

b. No Solution

$$x_1 - x_2 + x_3 = 1$$

c. Infinite no. of Solution.

Soln:

$B \neq 0$. So, non homogeneous linear system. d. Only one Solution.

$$[A:B] = \begin{bmatrix} 1 & 1 & 1 & 3 \\ 1 & -1 & 0 & 0 \\ 1 & -1 & 1 & 1 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1}} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -2 & -1 & -3 \\ 0 & -2 & 0 & -2 \end{bmatrix}$$

$$\xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & 1 & 1 & 3 \\ 0 & -2 & -1 & -3 \\ 0 & 0 & 1 & +1 \end{bmatrix} \quad \begin{array}{l} \text{Rank}(A) = 3 \\ \text{Rank}(A:B) = 3 \\ n = 3. \end{array}$$

$\therefore \text{Rank}(A) = \text{Rank}(A:B) = n \therefore \text{Unique Solution.}$

Q. Solve the System : $2x + 3y + z = 9$, $4x + y = 7$, $x - 3y - 7z = 6$.

Soln:

$$\begin{bmatrix} 2 & 3 & 1 & 9 \\ 4 & 1 & 0 & 7 \\ 1 & -3 & -7 & 6 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 2 & 3 & 1 & 9 \\ 0 & -5 & -2 & -11 \\ 1 & -3 & -7 & 6 \end{bmatrix} \xrightarrow{R_3 \rightarrow 2R_3 - R_1} \begin{bmatrix} 2 & 3 & 1 & 9 \\ 0 & -5 & -2 & -11 \\ 0 & -9 & -15 & 3 \end{bmatrix}$$

$$D = \begin{bmatrix} 2 & 3 & 1 \\ 4 & 1 & 0 \\ 1 & -3 & -7 \end{bmatrix} \Rightarrow \begin{array}{l} 2(-7) - 3(-28) + 1(-12-1) \\ -14 + 84 - 13 \\ 84 - 27 \Rightarrow 57. \end{array}$$

$$D_1 = \begin{bmatrix} 9 & 3 & 1 \\ 7 & 1 & 0 \\ 6 & -3 & -7 \end{bmatrix} \Rightarrow \begin{array}{l} -7(-21+3) + 1(-63-6) \\ -7(-18) + 1(-69) \\ 126 - 69 \Rightarrow 57. \end{array}$$

$$D_2 = \begin{bmatrix} 2 & 9 & 1 \\ 4 & 7 & 0 \\ 1 & 6 & -7 \end{bmatrix} \rightarrow \begin{aligned} &-4(-63-6) + 7(-14-1) \\ &-4(-69) + 7(-15) \\ &276 - 105 = 171 \end{aligned}$$

$$D_3 = \begin{bmatrix} 2 & 3 & 9 \\ 4 & 1 & 7 \\ 1 & -3 & 6 \end{bmatrix} \rightarrow \begin{aligned} &2(6+21) - 4(18+27) + 1(21-9) \\ &54 + 12 - 4(45) \\ &66 - 180 = -114 \end{aligned}$$

$$x = \frac{D_1}{D} = \frac{57}{57} = 1 \quad \rightarrow \quad \boxed{x=1}$$

$$y = \frac{D_2}{D} = \frac{171}{57} = 3 \quad \rightarrow \quad \boxed{y=3}$$

$$z = \frac{D_3}{D} = \frac{-114}{57} = -2 \quad \rightarrow \quad \boxed{z=-2}$$

Eigen Values and Eigen Vectors —

Let A be a square Matrix

$\lambda \rightarrow$ Eigen Value (characteristic root)

$X \rightarrow$ Eigen Vector (Non-Zero Vector)

$\rightarrow$ (characteristic Vector).

$$\boxed{AX = \lambda X} \quad ***$$

• Characteristic Equation —

$$AX = \lambda X$$

$$AX = \lambda IX$$

$$(AX - \lambda IX) = 0$$

$$(A - \lambda I)X = 0$$

(Infinite) $X \neq 0$ because

X is non zero Vector

$$\boxed{|A - \lambda I| = 0} \quad ***$$

This is characteristic Equation

- Eigen Value $\longrightarrow$ Root of the characteristic Equation.
 → Eigen Vector $\longrightarrow$ Reduced Row Echelon form.

Q. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$. Find the Eigen Value and Eigen Vector.

Soln:

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} -\lambda & 0 \\ 0 & -\lambda \end{bmatrix} = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(4-\lambda) - 4 = 0$$

$$4 - 5\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 5\lambda = 0$$

$$\lambda(\lambda - 5) = 0 \quad \text{then, } \boxed{\lambda = 0} \text{ or } \boxed{\lambda = 5}.$$

Eigen Values.

At $\lambda = 0$:-

$$\begin{bmatrix} 1-0 & 2 \\ 2 & 4-0 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \rightarrow \text{Square Matrix}$$

$$|A - \lambda I| \rightarrow |A| = 4 - 4 = 0 \quad (\text{Infinite Soln}).$$

$$R_2 \rightarrow R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Reduced Row Echelon form.}$$

Basic Variable $= x$

Free Variable $= y = t$

$$x + 2y = 0$$

$$x = -2y = -2t$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -2t \\ t \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

a. $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$

b. $\begin{bmatrix} -1 \\ 1/2 \end{bmatrix}$

c. $\begin{bmatrix} -4 \\ 2 \end{bmatrix}$

d. $\begin{bmatrix} 2 \\ -1 \end{bmatrix}$

Eigen Vector

corresponding to $\lambda = 0$.

At $\lambda = 5$:-

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \xrightarrow{R_2 \rightarrow 2R_2 + R_1} \begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1/2 \\ 0 & 0 \end{bmatrix} = \text{Reduced Row Echelon form}$$

Basic Variable = x Free Variable = $y = t$

$$x - 1/2 y = 0$$

$$x = 1/2 y = 1/2 t$$

$$X = \begin{bmatrix} 1/2 t \\ t \end{bmatrix} = t \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

Eigen Vector
Corresponding
to $\lambda = 5$ Q. If $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$. Find the Eigen Value of and Eigen Vectors.

Soln:

$$|A - \lambda I| = 0$$

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} -\lambda & 0 & 0 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{bmatrix} = 0$$

$$\begin{vmatrix} -\lambda & 0 & 1 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = 0$$

$$-\lambda (\lambda^2 - 0) + 1(0 + 0) = 0$$

$$-\lambda^3 = 0$$

$$\lambda = 0, 0, 0$$

 $\lambda = 0 \rightarrow$ Eigen Value of algebraic multiplicity = 3.• At $\lambda = 0$:-

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{\text{Reduced Row Echelon form.}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Basic Variables = z Free Variables = x and y .

$$x = t \text{ and } y = 1$$

$$z = 0$$

Eigen Vectors.

$$X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} t \\ 1 \\ 0 \end{bmatrix} = t \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$\lambda = 0$ is Eigen Value of Geometric multiplicity = 2.

Important Results -

1. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} = 0$$

Trace of the matrix = $1+4=5$.Sum of the Eigen Value = $0+5=5$.

$$(1-\lambda)(4-\lambda) - 4 = 0$$

$$4 - \lambda - 4\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 5\lambda = 0$$

$$\lambda(\lambda - 5) = 0$$

$$\lambda = 0, 5$$

Trace of the Matrix = Sum of Eigen Value.

2. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

$$|A| = \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} = 0$$

Determinant of the Matrix = Product of Eigen Value.

Eigen Value = 0, 5

Product of Eigen Value = 0

3. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

$$|A| = 4 - 4 = 0$$

If $|A| = 0$ then at least one eigen value is zero.

Product of Eigen Value = $\lambda_1 \lambda_2 = 0$

$$4. A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$1 + \lambda^2 - 2\lambda - 1 = 0$$

$$\lambda(\lambda - 2) = 0$$

$$\lambda = 0, 2$$

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 1 & 1 \\ 1 & 1-\lambda & 1 \\ 1 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$\lambda = 0, 0, 3$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

$$\lambda = 4, 0, 0, 0$$

$$A = [a_{ij}]_{n \times n} = 1 \quad \forall i \text{ and } j$$

↓ then

$$\text{Eigen value} = n, 0, 0, 0, \dots$$

5. Eigen Vectors corresponding to different Eigen Values are linearly independent.

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \lambda_1 = 0 \text{ and } \lambda_2 = 5.$$

$$\lambda_1 = 0 \Rightarrow$$

$$t \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 5 \Rightarrow$$

$$t \begin{bmatrix} 1/2 \\ 1 \end{bmatrix}$$

They are

linearly independent

On Multiplying -

not same so they are L.I.

6. Symmetric matrix :-

$$A = A^T$$

$$[a_{ij}] = [a_{ji}]$$

$$\text{Eg:- } \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$$

→ Symmetric Matrix.

$$|A - \lambda I| = \begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - 4 = 0$$

$$1 + \lambda^2 - 2\lambda + 4 = 0$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$\lambda^2 - 3\lambda + \lambda - 3 = 0$$

$$\lambda(\lambda - 3) + 1(\lambda - 3) = 0$$

$$(\lambda - 3)(\lambda + 1) = 0$$

$\therefore \lambda = 3, -1$ (Eigen Value).

• At $\lambda = -1$: $R_2 \rightarrow R_2 - R_1$

$$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 2 \\ 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1/2} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Reduced Row Echelon form.}$$

Basic Variables = x

free Variables = $y = t$

$$x + y = 0$$

$$x = -y = -t$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -t \\ t \end{bmatrix} = t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

↘ Eigen Vectors.

• At $\lambda = 3$,

$$\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \Rightarrow \begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow -R_1/2} \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Reduced Row Echelon form.}$$

Basic Variables = x

free Variables = $y = t$

$$x - y = 0$$

$$x = y = t$$

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

↘ Eigen Vectors.

Dot Product of Both Eigen Vectors = $1 \times (-1) + 1 \times 1 = -1 + 1 = 0$. ***

These Eigen Vectors are Orthogonal.

"Eigen Vector corresponding to different Eigen Values of Symmetric Matrix are orthogonal"

7. $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} \rightarrow$ Symmetric Matrix.

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - 4 = 0$$

$$\lambda^2 + 1 - 2\lambda - 4 = 0$$

$$\lambda^2 - 2\lambda - 3 = 0$$

$$\lambda^2 - 3\lambda + \lambda - 3 = 0$$

$$\lambda(\lambda - 3) + 1(\lambda - 3) = 0$$

$$(\lambda - 3)(\lambda + 1) = 0$$

$$\boxed{\lambda = 3, -1} \rightarrow \text{Real number.}$$

Eigen Value of Symmetric matrix or Hermitian are real number.

8. $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \rightarrow$ Skew Symmetric Matrix.

$$|A - \lambda I| = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda^2 = -1$$

$$\boxed{\lambda = \pm i}$$

Eigen Value of Skew Symmetric matrix or Skew Hermitian matrix are either purely imaginary or zero.

9. $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \rightarrow$ Orthogonal Matrix.

$$\boxed{\lambda = \pm i}$$

$$\boxed{|\lambda| = 1}$$

$$\text{If } z = a + ib \\ |z| = \sqrt{a^2 + b^2}$$

Eigen Value of unitary (orthogonal) matrix have absolute value 1.

10. Let, λ is the Eigen Value of matrix A then —

Matrix $\longrightarrow$ Eigen Value.

1. $KA \longrightarrow KA$

Eg: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

$5A = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix}$

Eigen Value of Matrix $A = 0,5$.

$5A = \begin{vmatrix} 5-\lambda & 10 \\ 10 & 20-\lambda \end{vmatrix} = 0$

$(5-\lambda)(20-\lambda) - 100 = 0$

$100 - 5\lambda - 20\lambda + \lambda^2 - 100 = 0$

$\lambda^2 - 25\lambda = 0$

$\lambda(\lambda - 25) = 0$

$\lambda = 0,25$

Eigen Value of Matrix $5A = 0,25$.

2. $A+KI \longrightarrow A+K$

$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

$\lambda_1 \quad \lambda_2$

$A \longrightarrow \begin{matrix} 0 & 5 \end{matrix}$

Eigen Value: $\lambda = 0,5$

$A+KI \longrightarrow A+K$

$A+2I = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$

$\uparrow$
2

$0+2 \quad 5+2$
 $\begin{bmatrix} 2 & 7 \end{bmatrix}$

$A+2I = \begin{bmatrix} 3 & 2 \\ 2 & 6 \end{bmatrix}$

Eigen Value $= \begin{vmatrix} 3-\lambda & 2 \\ 2 & 6-\lambda \end{vmatrix} = 0$

$(3-\lambda)(6-\lambda) - 4 = 0$

$18 - 9\lambda + \lambda^2 - 4 = 0$

$\lambda^2 - 9\lambda + 14 = 0$

$\lambda^2 - 7\lambda - 2\lambda + 14 = 0$

$\lambda(\lambda - 7) - 2(\lambda - 7) = 0$

$\lambda = 2,7$

3. $A^2 \longrightarrow \lambda^2$

Example: $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

Eigen Value of Matrix $A = 0,5$.

$$A^2 = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix}$$

$$\text{Eigen Value :- } \begin{bmatrix} 5-\lambda & 10 \\ 10 & 20-\lambda \end{bmatrix} = 0$$

$$(5-\lambda)(20-\lambda) - 100 = 0$$

$$100 - 25\lambda + \lambda^2 - 100 = 0$$

$$\lambda^2 - 25\lambda = 0$$

$$\lambda(\lambda - 25) = 0$$

$$\lambda = 0, 25$$

$$\text{Eigen Value of } A^2 \text{ Matrix} = 0^2, 5^2 = 0, 25.$$

4. $A \longrightarrow \lambda$
 $A^m \longrightarrow \lambda^m$ (m is positive integer).

5. $A \longrightarrow \lambda$ ***
 then Eigen Value of $\boxed{q_0 I + q_1 A + q_2 A^2 = q_0 + q_1 \lambda + q_2 \lambda^2}$

6. $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$

$$|A| = 0$$

Non invertible Matrix.

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$|A| = 4 - 6 = -2 \neq 0$$

Invertible Matrix.

$$\begin{bmatrix} -2-\lambda & 1 \\ 3/2 & -1/2-\lambda \end{bmatrix} = 0$$

$$-2(-0.5-\lambda) - 1(-0.5-\lambda) - 1.5 = 0$$

$$1 + 2\lambda + \frac{1}{2} + \lambda^2 - \frac{3}{2} = 0$$

$$\lambda^2 + 2\lambda + \frac{\lambda}{2} - \frac{1}{2} = 0$$

$$2\lambda^2 + 4\lambda + \lambda - 1 = 0$$

$$2\lambda^2 + 5\lambda - 1 = 0$$

$$\lambda = \lambda_3, \lambda_4$$

$$A^{-1} = \frac{1}{-2} \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\text{Eigen Value of Matrix } A = \lambda_1, \lambda_2$$

$$\boxed{\lambda_3 = \frac{1}{\lambda_1}, \lambda_4 = \frac{1}{\lambda_2}} \quad ***$$

Summary: $A \rightarrow$ Invertible Matrix $(|A| \neq 0) \rightarrow$ Eigen Value λ

then $A^{-1} \rightarrow \frac{1}{\lambda}$

7. $A \rightarrow \lambda$ adjoint $A = \frac{|A|}{\lambda}$ ***

$$A^{-1} = \frac{\text{adj} A}{|A|}$$

$$\text{adj} = |A| \cdot A^{-1}$$

adjoint $A = \frac{|A|}{\lambda}$

★ Eigen Value and Eigen Vector

• Eigen Value Given

• Matrix Given

Eigen Value of Matrix are given $\rightarrow$

Let 1, -1, 2 are Eigen Value of matrix A.

1. Trace of matrix A = Sum of eigen Value $= 1 - 1 + 2 = 2$.

2. Determinant of matrix A = Product of Eigen Values
 $= \lambda_1 \times \lambda_2 \times \lambda_3 = 1 \times (-1) \times 2 = -2$.

3. Eigen Value of $[2A]$ matrix = $2 \times 1, 2 \times (-1), 2 \times 2$
 $= 2, -2, 4$.

a. Determinant of $2A$ = Product of Eigen Value of $2A$
 $= 2 [1 \times -1 \times 2] = 2 \times (-2) \times 4 = -16$.

b. Trace of matrix $2A$ = Sum of Eigen Value of $2A$
 $= 2 - 2 + 4 = 4$.

4. Eigen Value of $A + 2I = 1 + 2$
 $= 1 + 2, -1 + 2, 2 + 2$
 $= 3, 1, 4$.

5. Eigen Value of $A^2 = \lambda^2$
 $= (1)^2, (-1)^2, (2)^2 = 1, 1, 4$

6. Eigen Value of $2I + 3A + A^2 =$

$$\lambda_1 = 2 + 3 \times 1 + 1^2 = 2 + 3 + 1 = 6.$$

$$\lambda_2 = 2 + 3(-1) + (-1)^2 = 2 - 3 + 1 = 0$$

$$\lambda_3 = 2 + 3(2) + (2)^2 = 2 + 6 + 4 = 12.$$

7. Eigen Value of $A^{-1} = \frac{1}{\lambda} \left\{ \frac{1}{+1}, \frac{1}{-1}, \frac{1}{\frac{1}{2}} \right\} = 1, -1, 1/2$

Determinant of $A^{-1} = \text{Product of Eigen Values of } A^{-1}$
 $= 1 \times (-1) \times \frac{1}{2} = -1/2.$

8. Eigen Value of adjoint $A = |A| \cdot \{ |A| = -2A \}$

$$\left\{ \frac{-2}{1}, \frac{-2}{-1}, \frac{-2}{\frac{1}{2}} \right\} = \{-2, 2, -1\}.$$

Important

9. Eigen Value of $(A - 2I)^{-1}$

Eigen Value of $A \rightarrow 1, -1, 2$

Eigen Value of $A - 2I \rightarrow 1 - 2, -1 - 2, 2 - 2$

Eigen Value of $A - 2I \rightarrow -1, -3, 0$

$|A - 2I| = 0 \rightarrow \text{non invertible}$

$(A - 2I)^{-1}$ does not exist.

10. Find the $|A^{100} + I| = ??$

$A \rightarrow 1, -1, 2$

$A^{100} \rightarrow 1, 1, 2^{100}$

$A^{100} + I \rightarrow 2, 2, 2^{100} + 1$

$|A^{100} + I| = 2 \times 2 \times (2^{100} + 1) \text{ Ans.}$

Matrix is given -

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

1. Eigen Value of the Matrix.

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 2 & 4-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(4-\lambda) - 4 = 0$$

$$4 - \lambda - 4\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 5\lambda = 0$$

$$\lambda(\lambda - 5) = 0$$

$$\boxed{\lambda = 0, 5}$$

2. Eigen Vector corresponding to Eigen Value λ_1 .

$$X = t \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$AX = \lambda_1 X$$

Matrix

Eigen Vector

Eigen Value.

Ex 3.

		Eigen Value	Eigen Vector
A	$\longrightarrow$	λ	X
A^m	$\longrightarrow$	λ^m	X

$$AX = \lambda X$$

$$XA = \lambda X$$

$$A^2 X = \lambda(A X) = \lambda^2 X$$

Multiply Sign

$$A^2 X = \lambda^2 X$$

$$XA = \lambda X$$

$$A^3 X = \lambda^2(A X)$$

$$\boxed{A^3 X = \lambda^3 X}$$

Important

$$\boxed{A^m X = \lambda^m X}$$

Caley-Hamilton Theorem -

- For Calculating higher Power of (A^n) Matrix in terms of A .
- For Calculating A^{-1} (negative Power) of the matrix in terms of A .

"Every Square Matrix Satisfies their characteristic Equation".

Ex: $A = \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix}$.

characteristic Eqn: $|A - \lambda I| = 0$

$$\left| \begin{bmatrix} 1 & 2 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} 1-\lambda & 2 \\ 3 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 - 6 = 0$$

$$\lambda^2 + 1 - 2\lambda - 6 = 0$$

$$\lambda^2 - 2\lambda - 5 = 0$$

By Caley-Hamilton Theorem put $\lambda = A$ in characteristic Equation

$$A^2 - 2A - 5I = 0$$

$$A^2 = 2A + 5I$$

$$\times A \quad \times A$$

$$A^3 = 2A^2 + 5AI$$

$$A^3 = 2[2A + 5I] + 5AI$$

$$A^3 = 4A + 10I + 5AI$$

$$A^3 = 9A + 10I$$

$$\times A \quad \times A$$

$$A^4 = 9A^2 + 10AI$$

$$A^4 = 9[2A + 5I] + 10AI$$

$$A^4 = 18A + 45I + 10AI$$

$$A^4 = 28A + 45I$$

• Negative Power of A —

$$A^2 - 2I - 5I = 0$$

$$\times A^{-1} \quad A^{-1}$$

$$A^{-1}(A^2 - 2A - 5I) = 0$$

$$A - 2I - 5A^{-1} = 0$$

$$5A^{-1} = A - 2I$$

$$A^{-1} = \frac{1}{5}(A - 2I)$$

Trick — { "Question and options are given in terms of A"
 "Higher Power of A"
 "Equations in terms of A"

Always use Cayley Hamilton Theorem.

— x — x — x — x — x —

1	0	0	0	1
2	0	0	1	001
3	0	1	000	001
4	1	000	000	001